MBA OPTIMIZATION TEACHING CASE

Meridian's Friday Allocation

When 50,000 vaccine doses must answer 83,500 requests

Student case and modeling assignment

50,000 doses   83,500 requested   8 counties

All organizations, locations, policy facts, and numerical data are fictional.

The decision

At 4:10 p.m. on Thursday, Dr. Lena Ortiz, operations director for the fictional Meridian Public Health Authority, received the final depot inventory report. By noon Friday, she must recommend how many vaccine doses each depot should ship to each of eight counties for the following week.

The state has 50,000 doses. County requests total 83,500. The governor's office has asked for a plan that is practical, equitable, and directs limited doses where they will do the most good. No numerical priority among those goals has been provided.

The central modeling problem: Every dose sent to one county is a dose that cannot be sent somewhere else. The allocation rule determines who waits.

Operating context

The vaccine is treated as a one-dose intervention for this one-week planning exercise. Doses arrive at three depots on Monday and expire after the planning week. Depot inventories cannot be repositioned before allocation. Counties may receive shipments from more than one depot unless the analyst assumes otherwise.

The authority estimates a cold-chain reliability for each depot-to-county lane. A reliability of 0.970 means that planners expect 97.0 percent of shipped doses to remain usable. These estimates are uncertain and are not guarantees.

County information

County requests reflect local submissions, not verified clinical need. Administration capacity is the maximum number of usable doses that can be administered during the week. Priority-population counts are local estimates.

County	Request	Admin.	Priority	Vuln.	Risk	Rural
Alder	12,000	11,000	6,200	0.82	1.25	Yes
Benton	10,000	9,500	4,100	0.43	0.90	No
Clearwater	9,000	7,200	4,800	0.76	1.10	Yes
Dover	14,000	13,000	5,000	0.35	0.80	No
Easton	8,500	7,500	4,700	0.88	1.30	No
Franklin	11,500	10,000	3,800	0.58	0.95	No
Grove	7,500	6,500	4,200	0.91	1.35	Yes
Harbor	11,000	9,000	4,300	0.49	1.00	No

The vulnerability index and severe-outcome risk multiplier are relative planning measures. They are not probabilities, clinical diagnoses, or directives about how they must enter a model.

Depot supply

Depot	Available doses	Planning fact
North Depot	20,000	Inventory expires after the planning week.
Central Depot	17,000	Inventory expires after the planning week.
Coastal Depot	13,000	Inventory expires after the planning week.

Transportation lanes

The data workbook contains all 24 lane records. The compact exhibit below reports travel hours, cost per 1,000 shipped doses, and reliability.

Depot	County	Hours	Cost/1,000	Reliability
North	Alder	1.2	620	97.0%
North	Benton	0.8	480	97.5%
North	Clearwater	3.1	1,040	95.5%
North	Dover	1.4	690	97.0%
North	Easton	3.5	1,160	95.0%
North	Franklin	2.4	870	96.5%
North	Grove	4.0	1,280	94.0%
North	Harbor	3.2	1,080	95.5%
Central	Alder	2.8	940	98.5%
Central	Benton	2.4	820	99.0%
Central	Clearwater	1.0	520	97.5%
Central	Dover	2.2	760	98.5%
Central	Easton	0.9	470	98.0%
Central	Franklin	2.7	900	97.0%
Central	Grove	1.4	590	97.8%
Central	Harbor	3.1	1,020	96.5%
Coastal	Alder	3.4	1,110	99.0%
Coastal	Benton	2.9	980	98.5%
Coastal	Clearwater	3.0	1,030	98.5%
Coastal	Dover	1.7	720	97.5%
Coastal	Easton	3.2	1,090	99.0%
Coastal	Franklin	1.1	510	98.5%
Coastal	Grove	2.8	960	99.0%
Coastal	Harbor	0.8	450	98.0%

What management has not resolved

- Should the objective emphasize expected usable doses, health benefit, the worst-served county, transportation cost, or some combination?
- Does equity mean equal doses, equal percentages, minimum floors, priority-population protection, or preference for high-vulnerability counties?
- Should reliability be an expected yield, hard standard, penalty, or scenario?
- Must all expiring supply ship, or should the authority hold a reserve?
- Are shipments continuous or restricted to 500- or 1,000-dose pallets?
- What denominator should define fairness: requests, administration capacity, priority population, or something else?

Assignment

1. State and justify the assumptions needed to complete the decision model.
2. Define the decision variables, objective, and constraints in plain English and mathematically.
3. Solve the model and recommend a depot-to-county allocation.

4. Report expected usable doses, expected losses, cost, county fulfillment, minimum fulfillment, and high-vulnerability fulfillment.
5. Solve at least one materially different interpretation of equity or impact and explain the change.
6. Test assumptions that could reverse the decision, including supply, floors, cutoffs, cold-chain treatment, and granularity.

Submission standard: A strong analysis makes value judgments visible. A technically correct model with unexplained priorities is incomplete.

Use `Meridian_Vaccine_Data.xlsx` for the complete exhibits. All data are fictional and for classroom use only.